

ERASTEEL TECHNICAL DATASHEET: ASP 2030

Application Family: Cutting tools | Standards: AISI M35 PM, W.-Nr. 1.3244, HS 6-5-3-8

1. Grade Overview & Characteristics

ASP 2030 is a cobalt-alloyed powder metallurgy high-speed steel with superior hot hardness, excellent temper resistance, and high compressive strength. Recommended for demanding cutting operations on hard or heat-resistant alloys.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	1.25 - 1.31	4.0 - 4.4	4.8 - 5.2	6.2 - 6.6	3.0 - 3.2	8.2 - 8.8	None

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	65.0 - 67.0 HRC	Machinability	5.5/10
Max Reachable Hardness	67.0 HRC	Wear Resistance	8.0/10
Annealed Delivery State	Max 280-350 HB	Toughness	6.0/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	8.5/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	5.5/10

4. Industrial Applications

Gear hobs, broaches, heavy-duty end mills, cold heading dies, milling cutters, reamers, taps for high-strength steels.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1150-1200°C. Triple tempering at 560°C for 1 hour each. Max reachable working hardness 67 HRC.